Reference Notes

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Contents

Part I

Readings & Literature Review

1. Age-Related Macular Degeneration: A Review

Overview

Age-related macular degeneration (AMD) is a progressive degenerative disease of the macula, the central retinal region responsible for high-acuity vision. AMD is among the leading causes of irreversible vision loss in adults over age 55 and is projected to affect nearly 288 million individuals worldwide by 2040.

The disease primarily affects photoreceptors, retinal pigment epithelium (RPE), bruch membrane, and choriocapillaris. AMD progression involves degeneration of these structures, ultimately leading to central vision loss. Early and intermediate disease may be asymptomatic, while advanced disease can substantially impair reading, driving, facial recognition, and other daily activities.

Pathophysiology

AMD is considered a multifactorial disease driven by interactions among aging, genetic susceptibility, environmental exposures, inflammation, and metabolic dysfunction.

Major biological processes implicated include chronic inflammation, complement pathway dysregulation, oxidative stress, lipid and lipoprotein accumulation, impaired extracellular matrix maintenance and choriocapillaris degeneration. Normal aging causes reduced photoreceptor density lipid deposition within Bruch membrane reduced density of choriocapillaris vessels and increased vascular resistance

These alterations disrupt retinal homeostasis and contribute to extracellular deposit formation. The hallmark lesion of AMD is the druse (plural: drusen), consisting of lipids, proteins, and minerals accumulating between the RPE and Bruch membrane.

Classification of AMD

Stage	Defined By	Clinical Characteristics	OCT Findings	Notes
Early AMD	Medium drusen ($> 63\,\mu m$ and $\leq 125\,\mu m$) No AMD-associated pigmentary abnormalities	Usually asymptomatic Visual acuity often preserved	Small dome-shaped RPE elevations corresponding to drusen Relatively preserved retinal architecture	Progression risk is relatively low at this stage.
Intermediate AMD	Large drusen ($> 125\,\mu m$) And/or pigmentary abnormalities	Difficulty in low-light conditions Reduced contrast sensitivity Metamorphopsia Mild visual blurring	Larger drusen elevations Hyperreflective foci Pigment migration Structural irregularities of the RPE	Most important stage for prediction studies because progression to advanced disease frequently occurs from this state.
Late AMD: Geographic Atrophy	Confluent regions of photoreceptor loss RPE loss Choriocapillaris degeneration	Progressive central vision loss Central scotomas Declining visual acuity	Loss of outer retinal layers RPE attenuation or absence Increased signal transmission into the choroid Hypertransmission defects	Geographic atrophy enlarges over time and is associated with irreversible visual loss.
Late AMD: Neovascular	Growth of abnormal blood vessels beneath or within the retina VEGF-mediated angiogenesis	Often rapid symptom onset Sudden vision decline Metamorphopsia Central visual defects	Intraretinal fluid Subretinal fluid Pigment epithelial detachments Hemorrhage-associated abnormalities	Responsible for most severe acute vision loss associated with AMD.

Table 1: Clinical classification of age-related macular degeneration (AMD).

Major Risk Factors

Age

Age is the strongest risk factor.

The incidence of late AMD increases dramatically with age:

- 0.3 per 1000 individuals aged 55–59
- 5.7 per 1000 individuals aged 75–79
- 36.7 per 1000 individuals aged 90+

Genetics

Estimated heritability is approximately 71%.

Major genetic loci include CFH and ARMS2-HTRA1, which are involved in complement regulation, inflammation, lipid metabolism, and extracellular matrix maintenance.

Smoking

Smoking is the strongest established environmental risk factor. Studies consistently demonstrate substantially increased AMD incidence among heavy smokers.

Diet and Lifestyle

Protective factors include fish consumption, fruits and veggies, and physical activity.

Features Relevant to Progression Prediction

For a progression-prediction project, the review repeatedly highlights several structural features associated with worsening disease.

Drusen Burden

Important characteristics include the number of drusen, total area, volume, and presence of large drusen. Increasing drusen burden characterizes progression from early to intermediate AMD.

Pigmentary Abnormalities

Pigment changes arise from migration of RPE cells into inner retinal layers.

These appear as:

- Hyperpigmentation on fundus imaging
- Hyperreflective foci on OCT

The review identifies pigmentary abnormalities as a major marker of progression risk.

Hyperreflective Foci

Migration of RPE cells creates hyperreflective structures visible on OCT.

These likely reflect active retinal remodeling and degeneration.

Outer Retinal Degeneration

Progressive loss of photoreceptors, ellipsoid zone integrity, and RPE structure may preced advanced disease development.

Hypertransmission Defects

Loss or attenuation of the RPE permits increased OCT signal penetration into deeper tissue.

These appear as vertical transmission columns extending into the choroid.

Such findings are particularly relevant to the current project because the proposed "barcoding" phenomenon appears visually related to localized transmission abnormalities beneath regions of RPE degeneration.

Geographic Atrophy Precursors

Potential indicators of future atrophy include RPE disruption, hypertransmission, outer retinal thinning, or progressive drusen remodeling.

High-Risk Clinical Profile

The AREDS study reported that:

- 25.9% of individuals with large drusen and pigment abnormalities progressed to late AMD within 5 years
- If one eye already had advanced AMD, progression in the fellow eye reached 53.1% within 5 years when large drusen and pigment abnormalities were present

These findings make large drusen and pigmentary abnormalities particularly important baseline predictors.

2. The Barcode Sign in Optical Coherence Tomography

Overview

El Zawahry et al. described a characteristic OCT imaging pattern termed the *barcode sign*, observed in eyes with extensive corneal neovascularization. The barcode sign appears as multiple vertically oriented dark stripes extending through the OCT image. These stripes arise from attenuation of the OCT signal caused by structures that strongly block light transmission. Although the study focuses on corneal vascularization rather than age-related macular degeneration (AMD), it provides an important example of how biological structures can generate stripe-like OCT patterns through optical shadowing mechanisms.

Study Design

The authors examined five patients with extensive corneal neovascularization arising from:

- Infectious keratitis
- Limbal stem cell deficiency
- Interstitial keratitis

Anterior-segment OCT scans were obtained using a Heidelberg Spectralis OCT system.

For each patient:

- Twenty-one scans were acquired across the superior cornea
- Twenty-one scans were acquired across the inferior cornea
- Forty-two scans were analyzed per eye

OCT findings were compared against slit-lamp examination and clinical imaging.

Description of the Barcode Sign

Major vascular trunks within the cornea completely obstructed OCT light transmission.

This produced:

- A vertical hypo-reflective stripe
- Extension of the stripe posterior to the vessel
- Stripe width proportional to vessel width
- Multiple parallel stripes when several vessels were present

When many vessels were visualized simultaneously, the OCT image resembled a retail barcode pattern, leading to the term "Barcode Sign" where the stripes appeared as dark linear regions embedded within the brighter OCT background.

Proposed Mechanism

The authors hypothesized that the shadowing effect arises primarily from the blood column within actively perfused vessels rather than the vessel wall itself.

Evidence supporting this interpretation includes:

- Both afferent and efferent vessels generated equivalent shadows
- Shadow intensity did not appear dependent on vessel-wall structure
- Ghost vessels lacking active blood flow did not generate shadows

Therefore, the attenuation pattern appears linked to the presence of circulating blood. The barcode sign may thus represent an OCT manifestation of active vascular perfusion.

Relationship to OCT Physics

The barcode sign illustrates a general OCT principle: Structures that strongly attenuate or absorb OCT light can generate vertical shadow artifacts extending into deeper tissue layers.

The resulting pattern consists of:

- Localized signal loss
- Vertical transmission defects
- Stripe-like image structures
- Reduced visibility of deeper anatomy

Such shadowing artifacts are well known in OCT imaging and can reveal information about the underlying structure causing attenuation.

Relevance to AMD Progression Research

Although the barcode sign was identified in corneal vascularization, the study is conceptually relevant to AMD imaging. Several advanced AMD features can alter OCT signal transmission, including:

- Retinal pigment epithelium (RPE) degeneration
- Geographic atrophy
- Hypertransmission defects
- Choroidal structural abnormalities
- Neovascular complexes

These abnormalities can create vertically organized intensity patterns within OCT volumes. Consequently, the barcode sign provides evidence that biologically meaningful stripe-like structures may emerge naturally from tissue-specific optical interactions rather than image-processing artifacts.

Implications for the Current Project

The AMD progression project investigates whether stripe-like "barcoding" patterns visible within OCT scans are associated with future disease progression.

The barcode sign literature suggests several hypotheses:

- Stripe patterns may arise from localized attenuation or transmission phenomena.
- Stripe width may reflect properties of the underlying anatomical structure.
- Stripe density may provide information about disease burden.
- Quantitative characterization of stripe morphology may produce useful imaging biomarkers.

Potential quantitative descriptors include stripe count, spacing, width, contrast, orientation, or spatial frequency content.

While the biological mechanism in AMD may differ from corneal vascular shadowing, this study provides an important precedent demonstrating that barcode-like OCT patterns can arise from meaningful underlying tissue structure and physiology.

3. Self-Attention CNN for Retinal Layer Segmentation

Overview

Cao et al. proposed a hybrid convolutional neural network (CNN) and Transformer architecture for retinal layer segmentation in optical coherence tomography (OCT) images. The primary objective was to improve segmentation accuracy by combining the local feature extraction capabilities of CNNs with the long-range dependency modeling provided by self-attention mechanisms.

The proposed architecture extends the U-Net framework by incorporating:

- A Transformer-based self-attention module,
- Attention Gates (AGs),
- Channel attention mechanisms,
- A one-dimensional attention strategy tailored to retinal OCT anatomy.

The model was evaluated on public retinal OCT datasets and demonstrated improved segmentation performance compared to several existing CNN-based and Transformer-based approaches.

Motivation

Retinal OCT images possess highly structured anatomical organization. Traditional CNNs perform well at extracting local image features but have limited receptive fields, making it difficult to capture long-range spatial relationships.

Challenges in OCT segmentation include layer boundaris spanning large portions of the image, anatomical variability, pathology-induced distortion, and loss of contextual information during downsampling.

The authors argue that successful segmentation requires both local texture information and global structural context. Self-attention mechanisms are introduced to address this limitation.

Self-Attention Mechanism

The Transformer component allows image regions to interact with one another regardless of spatial distance. Self-attention is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V,$$

where $Q =$ query matrix, $K =$ key matrix, $V =$ value matrix, and $d =$ feature dimension.

Unlike convolution, which uses local neighborhoods, self-attention provides a global receptive field. This allows the model to learn relationships between distant retinal structures and improve segmentation consistency across the image.

One-Dimensional Transformer Design

A key innovation of the paper is the use of a one-dimensional Transformer. The authors observed that retinal anatomy is organized primarily along the vertical direction. Instead of computing attention across the entire two-dimensional image, feature maps are rearranged so that attention operates only along the retinal depth dimension.

Feature maps are reshaped from

$$(B, C, H, W) \rightarrow (BW, H, C)$$

where B = batch size, C = number of channels, H = image height, and W = image width.

This design preserves anatomical structure, reduces computational cost, increases effective receptive field, and improves efficiency relative to full 2D attention.

Encoder–Decoder Architecture

The overall architecture follows the encoder–decoder paradigm of U-Net.

Encoder

The encoder consists of repeated blocks of:

- 3×3 convolutions,
- Batch normalization,
- ReLU activation,
- Max pooling.

As depth increases, spatial resolution decreases and feature complexity increases. The encoder extracts increasingly abstract representations of retinal structures.

Decoder

The decoder performs:

- Upsampling,
- Feature fusion,
- Boundary reconstruction,
- Pixel-wise segmentation.

Skip connections transfer high-resolution information from encoder layers to corresponding decoder layers. These connections help preserve fine retinal boundary details that may otherwise be lost during downsampling.

Attention Gates

The model incorporates Attention Gates (AGs) within skip connections. Rather than passing all encoder features directly to the decoder, attention mechanisms determine which features are most relevant.

Conceptually:

$$\tilde{x} = \alpha x, \quad \forall \quad 0 \leq \alpha \leq 1$$

where α is a learned attention coefficient.

Benefits include regularization (suppression) of irrelevant features, improved focus on retinal boundaries, and enhanced segmentation precision.

Channel Attention

In addition to spatial attention, channel attention is used to weight feature channels according to importance. Different channels often capture different types of information:

- Edges,
- Layer boundaries,
- Texture patterns,
- Anatomical structures.

Channel attention allows the network to emphasize informative channels and reduce the influence of less useful features.

Loss Function

Training uses a weighted combination of Dice loss and multiclass cross-entropy loss.

The Dice coefficient is

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}.$$

Corresponding Dice loss L_{Dice} and cross-entropy loss L_{CE} are:

$$L_{\text{Dice}} = 1 - \text{Dice}. \quad L_{\text{CE}} = -\sum_i y_i \log(p_i).$$

The overall objective is

$$L = \lambda_1 L_{\text{Dice}} + \lambda_2 L_{\text{CE}}.$$

Combining the two losses balances pixel level calssification accuracy and global segmentation overlap.

Datasets

The model was evaluated on two retinal OCT datasets:

- DUKE Diabetic Macular Edema (DME) Dataset,
- Optic Disc Retinal OCT Dataset.

These datasets contain manually annotated retinal layer segmentations used as ground truth labels.

Results

Performance was evaluated using the Dice coefficient. For the DUKE DME dataset, an average score of 0.871 was reported, and for the optic-disc dataset, 0.820 was reported. This outperforms UNet, ATtention UNet, TransUNet, RelayNet, and DRUNey. The improvements suggest that combining CNN-based local feature extraction with Transformer-based global context modeling can improve retinal layer segmentation performance.

Key Takeaways

The primary contribution of this work is the integration of self-attention into a U-Net style segmentation framework for retinal OCT images.

Important observations include:

- CNNs effectively learn local retinal texture and boundary information.
- Self-attention provides global contextual information and long-range dependency modeling.
- Restricting attention to the vertical dimension leverages the layered anatomical structure of retinal OCT images.
- Attention Gates improve feature selection during decoding.
- Combining local and global representations improves segmentation accuracy relative to standard CNN architectures.

The paper demonstrates the potential value of hybrid CNN–Transformer architectures for retinal OCT analysis and segmentation tasks.

4. Machine Learning for Detection of Geographic Atrophy in Spectral OCT

This paper presents two deep learning pipelines for automated detection and quantification of geographic atrophy (GA) from spectral-domain optical coherence tomography (SD-OCT). Rather than relying on fundus autofluorescence (FAF), the authors use volumetric OCT data to identify regions of retinal pigment epithelium (RPE) loss and associated hypertransmission defects. Although the work focuses on supervised semantic segmentation, several of the ideas are directly applicable to our goal of automatically identifying and quantifying barcoding artifacts.

4.1. Clinical Motivation

Geographic atrophy represents the advanced stage of dry age-related macular degeneration (AMD), characterized by degeneration of the retinal pigment epithelium (RPE), photoreceptors, and outer retinal layers. Progression to GA is irreversible and results in permanent vision loss.

The Classification of Atrophy Meetings (CAM) consensus defines complete RPE and outer retinal atrophy (cRORA) on OCT by the simultaneous presence of

- complete RPE loss,
- overlying photoreceptor degeneration, and
- hypertransmission into the choroid,

with each lesion spanning at least $250\,\mu$m in its greatest diameter.

The authors argue that OCT is preferable to fundus autofluorescence because it provides three-dimensional structural information and allows direct visualization of retinal layers rather than only surface appearance.

4.2. Ground Truth Construction

A particularly interesting aspect of the work is that the authors do not manually delineate every GA lesion from scratch. Instead, they leverage retinal layer segmentation.

Previously validated machine learning models segment three retinal boundaries:

- Ellipsoid Zone (EZ)
- Retinal Pigment Epithelium (RPE)
- Bruch's Membrane (BM)

Ground-truth GA masks are then generated automatically wherever these three segmented layers completely overlap, indicating total disappearance of both the EZ and RPE.

Conceptually,

$$\text{GA} \iff EZ = RPE = BM$$

within a local region.

This produces pixel-level training masks while dramatically reducing manual annotation effort.

4.3. B-Scan versus En Face Models

The paper develops two complementary segmentation models.

The first operates directly on individual OCT B-scans. This approach detects GA using cross-sectional structural information, making it highly specific to retinal anatomy. Because every slice is analyzed independently, subtle structural changes are preserved.

The second operates on en face OCT projections. Instead of detecting retinal layer loss directly, the model identifies regions of hypertransmission visible in the en face image. Predictions can then be projected back onto individual B-scans using the Bruch's membrane segmentation.

The authors emphasize that the two representations provide complementary information:

- B-scans capture detailed retinal anatomy.
- En face images capture global lesion morphology.

Future work proposed by the authors is to combine both models into a hybrid system.

4.4. Model Architecture

Both models employ a standard U-Net architecture for semantic segmentation.

Training used

- resized 256×256 inputs,
- binary cross-entropy loss,
- approximately twenty million trainable parameters,
- automatic hard-example retraining by oversampling poorly performing patches.

Although technically straightforward, the authors demonstrate that careful construction of the training labels contributes more to performance than architectural novelty.

4.5. Results

The B-scan model achieved

- 91% detection accuracy,
- 94% pixel-wise segmentation accuracy,
- ICC = 0.88 for lesion measurements.

The en face model achieved

- 82% detection accuracy,
- 96% pixel-wise segmentation accuracy,
- ICC = 0.95 for lesion measurements.

Performance generalized well across both Heidelberg Spectralis and Zeiss Cirrus OCT systems, suggesting that the learned structural features are relatively device independent.

4.6. Hypertransmission as an Imaging Biomarker

Perhaps the most interesting observation is that many apparent false positives from the en face model were not actually incorrect predictions.

Instead, the model frequently identified isolated hypertransmission defects that did not yet satisfy the complete CAM definition of geographic atrophy.

Consequently, the authors hypothesize that hypertransmission itself may represent a useful biomarker preceding fully developed GA. Rather than suppressing these predictions, they suggest explicitly modeling hypertransmission independently from established GA.

This observation is particularly relevant because our project is similarly interested in identifying image features that may precede disease progression rather than only detecting already established pathology.

4.7. Relevance to Barcoding Detection

Although the objective differs substantially from ours, several ideas transfer directly.

First, the paper demonstrates that OCT contains sufficient structural information for automated discovery of clinically meaningful biomarkers.

Second, the authors emphasize representation learning directly from image structure rather than relying solely on handcrafted measurements.

Third, the work illustrates how clinically meaningful imaging biomarkers can be quantified automatically once reliable image representations have been obtained.

Our problem differs in one important respect. Rather than detecting a well-defined pathological structure such as GA, we seek to characterize vertical barcoding artifacts whose morphology is not currently described by a precise anatomical definition. Consequently, constructing exhaustive pixel-level annotations suitable for supervised segmentation would be difficult.

Instead, our objective is to develop an unsupervised representation capable of identifying and quantifying recurring vertical stripe patterns without requiring dense manual labels.

4.8. Limitations

From the perspective of the present project, several limitations remain.

- The approach requires extensive pixel-level ground truth.
- Performance depends upon accurate retinal layer segmentation.
- U-Net learns supervised anatomical labels rather than discovering novel biomarkers.
- The framework detects predefined pathology instead of identifying previously unknown structural patterns.

These limitations motivate investigating unsupervised feature learning approaches for quantitative characterization of barcoding.

4.9. Method Sketch

Inspired by the OCT-based workflow of Kalra et al., our approach similarly begins by leveraging the anatomical structure available within individual B-scans rather than attempting to analyze the entire OCT image directly. However, instead of performing supervised semantic segmentation of geographic atrophy, our objective is to discover and quantify barcoding patterns using unsupervised image analysis.

The proposed pipeline consists of four stages.

First, each Heidelberg Spectralis B-scan is paired with the retinal layer segmentations provided by the HEYEX software. These segmentations are used to identify the Bruch's membrane (BM), which serves as a stable anatomical reference. Each B-scan is flattened with respect to the BM in order to remove normal retinal curvature and standardize image geometry across subjects.

Following flattening, only the image region at and below the BM is retained. This isolates the portion of the OCT image where hypertransmission and barcoding are expected to occur while removing the majority of unrelated retinal structure. Restricting the analysis to this anatomically meaningful region substantially reduces background variation and focuses subsequent processing on tissue most relevant to the proposed biomarker.

Within this region of interest, an unsupervised algorithm will be used to identify candidate barcoding regions based on image appearance. Rather than relying upon manually defined annotations, the method will learn or estimate characteristic intensity patterns associated with barcoding. Candidate features include increased local brightness, oscillatory vertical intensity profiles, periodic texture, or combinations thereof. Multiple unsupervised approaches will be explored, and the final representation will be selected based on robustness and consistency across patients.

Instead of producing a dense pixel-level segmentation mask, the initial objective is to identify the horizontal extent of barcoding along each B-scan. The algorithm should therefore produce an overlay indicating regions classified as normal and regions exhibiting barcoding, for example,

$$\underbrace{\qquad\qquad\qquad}_{\text{Normal}}\;\;\underbrace{\qquad\qquad\qquad}_{\text{Barcoding}}\;\;\underbrace{\qquad\qquad}_{\text{Normal}}$$

where the highlighted interval corresponds to the automatically detected barcoding region.

Once the affected region has been localized, quantitative biomarkers can be extracted exclusively within the detected interval. Candidate measurements include the horizontal extent of barcoding, average and maximum brightness, vertical intensity oscillation statistics, texture descriptors, and additional quantitative measures that may correlate with disease progression. By separating localization from quantification, the framework naturally supports future refinement of both detection algorithms and downstream biomarkers without altering the overall pipeline.

Part II

Project Directions

5. Barcoding Detection and Quantification

The aim now is to figure out whether V_2 (Baseline) predicts Early Atrophy Progression or not. To do so, we will take the five fastest progressors and five slowest progressors and create a ground truth by marking EA and barcoding regions.

5.1. Data Preparation

We first take the five fastest progressors and five slowest progressors and create a ground truth by marking EA and barcoding regions. We will then use this ground truth to train a model to detect barcoding regions in new images.

1. Double click patient
2. Go to V2 HRA OCT and select the smallest scan (20x20)
3. Got to Display → Overlay → Measure Distance
4. Lines must be flat, mark the barcoding regions (RED) and EA regions (YELLOW)
5. Image → Export as picture (JPG) → Add Overlays

5.2. Future Work

Following creation of the ground truth dataset, several directions will be investigated.

First, the manually annotated images will be analyzed to understand the spatial relationship between Early Atrophy (EA) and barcoding regions, as well as differences between fast and slow progressors. EDA will guide the choice of modeling approach and help determine whether the annotations are sufficiently consistent for automation.

ML methods capable of localizing clinically relevant regions will be explored such as deep learning models for localization and semantic segmentation, weakly supervised methods that leverage the clinician annotations, and existing OCT analysis methods from the literature. The objective is to produce an interpretable overlay that highlights predicted EA and barcoding regions on previously unseen baseline OCT scans.

Once reliable localization has been achieved, quantitative measurements of barcoding will be investigated. Potential biomarkers include the horizontal extent of barcoding, the number of distinct barcode regions, total barcode coverage, overlap with Early Atrophy, and the spatial relationship between EA and barcoding.

Finally, the developed methods will be validated against the clinician-generated ground truth to assess localization accuracy, measurement consistency, and clinical interpretability.

6. Proposed Sliding-Window Barcoding Detection Framework

6.1. Overview

The proposed method is an interpretable, classical image-analysis approach for identifying horizontal regions of an OCT B-scan that exhibit barcode-like vertical transmission patterns. Rather than classifying individual image pixels, the method evaluates a sequence of overlapping vertical windows that span the full depth of the processed sub-Bruch's membrane (BM) region, like a convolutional layer

For each horizontal window, several mathematical features are calculated to describe:

1. the degree of vertical organization,
2. the repetition or periodicity of vertical structures,
3. the intensity or amplitude of hypertransmission,
4. the persistence of column-like structures with depth, and
5. the spatial heterogeneity of the image texture.

The features are standardized and combined into a single continuous barcode score. Horizontal positions whose score exceeds a selected threshold are marked as candidate barcoding regions. Short detections are removed and nearby detections may be merged to produce contiguous barcode intervals.

The complete workflow is:

$$\texttt{.E2E} \rightarrow \texttt{B-scan} \rightarrow \texttt{flattening} \rightarrow \texttt{sub-region crop} \rightarrow \texttt{normalization} \rightarrow \texttt{denoising} \rightarrow \texttt{feature engineering} \rightarrow \texttt{weighted score} \rightarrow \texttt{intervals}.$$

6.2. E2E Loading and B-Scan Selection

Each Heidelberg E2E file is imported as a volumetric OCT scan. If the analysis is intended to compare subjects, the central B-scan is selected separately for each volume using

$$i_{\text{center}} = \left\lfloor \frac{N_{\text{B-scans}}}{2} \right\rfloor,$$

where:

- $N_{\text{B-scans}}$ is the total number of B-scans in the volume,
- i_{center} is the zero-based index of the selected central scan, and
- $\lfloor \cdot \rfloor$ denotes rounding downward to the nearest integer.

This definition allows volumes containing different numbers of B-scans to be processed consistently. For within-volume validation, the same preprocessing configuration can instead be applied across all B-scans in one selected volume.

6.3. Flattening to Bruch's Membrane

The BM boundary is represented by one vertical coordinate for each horizontal image column: $b(x)$, where x is the horizontal column position, and $b(x)$ is the detected BM row at column x. A common reference row is selected, typically the median BM position:

$$b_{\text{ref}} = \text{median}_x \{b(x)\} .$$

Each column is shifted vertically by

$$\Delta(x) = b_{\text{ref}} - b(x).$$

The purpose of this operation is to transform the curved BM boundary into a horizontal line. After flattening, the same vertical coordinate represents approximately the same anatomical depth below BM across all horizontal positions.

Flattening is important because the proposed detector measures vertical persistence and orientation. Without flattening, natural retinal curvature could be incorrectly interpreted as directional or heterogeneous image structure.

6.4. Sub-BM Region of Interest

After flattening, a fixed-depth region is extracted beginning at BM:

$$I_{\text{ROI}} = I_{\text{flat}} \left[b_{\text{ref}} : b_{\text{ref}} + D - 1, \, :\right] ,$$

where:

- I_{flat} is the flattened B-scan,
- D is the selected depth below BM,
- I_{ROI} is the processed sub-BM region.

The initial configuration uses D=150 pixels so the resulting image is $\mathbb{R}^{150\times 512}$ with:

- rows representing depth below BM,
- columns representing horizontal retinal position.

6.5. Intensity Normalization

Intensity normalization is used to reduce variation caused by image brightness, contrast, and acquisition conditions. Two normalization families may be evaluated.

6.5.1. Whole-ROI Z-Score Normalization

For whole-ROI z-score normalization,

$$Z(z, x) = \frac{I_{\mathrm{ROI}}(z, x) - \mu_{\mathrm{ROI}}}{\sigma_{\mathrm{ROI}} + \varepsilon},$$

where:

- $I_{\mathrm{ROI}}(z, x)$ is the intensity at depth z and horizontal position x,
- μ_{ROI} is the mean intensity across all pixels in the sub-BM ROI,
- σ_{ROI} is the standard deviation of all ROI intensities,
- ε is a small positive constant that prevents division by zero,
- $Z(z, x)$ is the normalized intensity.

After z-score normalization:

- $Z = 0$ represents the average ROI intensity,
- $Z > 0$ represents brighter-than-average regions,
- $Z < 0$ represents darker-than-average regions.

Z-score normalization is useful when the detector combines several mathematical features whose scales should be comparable.

6.5.2. Whole-ROI Percentile Normalization

The previously tested percentile normalization may also be retained as an alternative configuration:

$$I_{\mathrm{norm}}(z, x) = \mathrm{clip}\left(\frac{I_{\mathrm{ROI}}(z, x) - q_\ell}{q_u - q_\ell}, 0, 1\right),$$

where:

- q_ℓ is the selected lower intensity percentile,
- q_u is the selected upper intensity percentile,
- $\mathrm{clip}(\cdot, 0, 1)$ restricts values to the interval $[0, 1]$.

Pixels below the first percentile are mapped to 0, pixels above the ninety-ninth percentile are mapped to 1, and intermediate values are linearly rescaled. The normalization method will therefore be treated as a configurable experimental parameter:

- z-score normalization,
- 1–99 percentile normalization,
- 0–100 min–max normalization,
- no normalization.

6.6. Two-Dimensional Gaussian Denoising

A mild Gaussian filter is applied before feature extraction to reduce speckle noise while preserving larger vertical structures. The smoothed image is

$$I_{\sigma_z,\sigma_x} = G_{\sigma_z,\sigma_x} * I_{\text{norm}},$$

where:

- G_{σ_z,σ_x} is a two-dimensional Gaussian kernel,
- $*$ denotes convolution,
- σ_z controls smoothing along depth,
- σ_x controls smoothing horizontally,
- I_{σ_z,σ_x} is the denoised image.

An anisotropic Gaussian configuration ($\sigma_z > \sigma_x$) may be preferable as it applies more smoothing along the vertical depth direction while applying less smoothing across horizontal columns. The purpose is to reduce speckle without excessively blurring narrow vertical barcode structures.

An initial configuration may be

$$\sigma_z = 1.0, \qquad \sigma_x = 0.5.$$

These values are not treated as final and should be included in sensitivity testing.

6.7. Sliding-Window Configuration

The detector operates on overlapping vertical windows that span the full ROI depth. For a window centered at horizontal position x, define

$$W_x = I_{\sigma_z,\sigma_x}\left[0 : D-1,\, x-r : x+r\right],$$

where:

- D is the ROI depth,
- r is the horizontal window radius,
- $2r+1$ is the window width,
- W_x is the image patch used to calculate features at position x.

For example, a 21-pixel window has

$$r = 10, \qquad 2r+1 = 21.$$

The first windows are therefore approximately (0,20), (1,21), and (2,22). Thus, the windows overlap and move horizontally by a configurable stride. *The initial configuration is a 21 pixel window with* *a 1 pixel stride.*

A stride of one produces one local score for approximately every horizontal position. Larger strides reduce computation but produce coarser localization. Reflection padding may be used near the left and right boundaries so that every column receives a full-width local window.

6.8. Feature 1: Vertical Orientation and Verticality

Barcode regions are expected to contain elongated structures that remain approximately vertical. Directional organization can be quantified using the structure tensor. Let

$$I_x = \frac{\partial I}{\partial x}, \qquad I_z = \frac{\partial I}{\partial z},$$

where:

- I_x is the horizontal image-intensity gradient,
- I_z is the vertical image-intensity gradient,
- a gradient measures how rapidly image brightness changes in a particular direction.

The local structure tensor is

$$J = \begin{bmatrix} \langle I_x^2 \rangle & \langle I_x I_z \rangle \\ \langle I_x I_z \rangle & \langle I_z^2 \rangle \end{bmatrix},$$

where:

- $\langle I_x^2 \rangle$ is the locally averaged squared horizontal gradient,
- $\langle I_z^2 \rangle$ is the locally averaged squared vertical gradient,
- $\langle I_x I_z \rangle$ represents interaction between the two directions,
- the local averaging reduces sensitivity to individual noisy pixels.

The structure tensor has two eigenvalues:$\lambda_1 \geq \lambda_2 \geq 0$. which summarize the strength of directional image variation. An anisotropy score is

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2 + \varepsilon}.$$

Its interpretation is:

- $A \approx 0$: no strong preferred orientation,
- $A \approx 1$: strongly directional or elongated structure.

The dominant local orientation may be estimated by

$$\theta = \frac{1}{2} \tan^{-1} \left(\frac{2J_{xz}}{J_{xx} - J_{zz}} \right),$$

where:

- $J_{xx} = \langle I_x^2 \rangle$,
- $J_{zz} = \langle I_z^2 \rangle$,
- $J_{xz} = \langle I_x I_z \rangle$,
- θ is the estimated dominant gradient orientation.

Because line orientation is perpendicular to gradient orientation, the implementation must explicitly convert the gradient direction into the corresponding structural direction. A verticality score for window W_x may be defined as

$$V(x) = \frac{1}{|W_x|} \sum_{(z,u)\in W_x} A(z,u)\,\mathbf{1}\left(\left|\theta_{\text{line}}(z,u) - \frac{\pi}{2}\right| < \delta_\theta\right),$$

where:

- $|W_x|$ is the number of pixels in the window,
- θ_{line} is the estimated line orientation,
- $\pi/2$ corresponds to a vertical line,
- δ_θ is the allowed angular tolerance,
- $\mathbf{1}(\cdot)$ equals 1 when the condition is true and 0 otherwise.

A high $V(x)$ means that a large proportion of the window contains strongly oriented vertical structure.

6.9. Feature 2: Column Persistence With Depth

A barcode-like structure should persist across a meaningful portion of the sub-BM depth rather than appearing as an isolated bright pixel. For each column u, define a binary candidate map:

$$B(z,u) = \mathbf{1}\left[C(z,u) > \tau_C\right],$$

where:

- $C(z,u)$ may represent intensity, verticality, or a combination of both,
- τ_C is the threshold used to define a candidate barcode pixel.

The depth fraction for one column is

$$P_{\text{fraction}}(u) = \frac{1}{D} \sum_{z=0}^{D-1} B(z,u).$$

This represents the proportion of the sub-BM depth containing candidate barcode signal. A connected-run persistence score may also be used:

$$P_{\text{run}}(u) = \frac{L_{\max}(u)}{D},$$

where $L_{\max}(u)$ is the longest consecutive vertical run of candidate pixels in column u. The window-level persistence score is then

$$P(x) = \frac{1}{2r+1} \sum_{u=x-r}^{x+r} P_{\text{run}}(u).$$

A high $P(x)$ indicates that vertical signals extend consistently through depth across several neighboring columns.

6.10. Feature 3: Horizontal Periodicity

Barcoding may contain repeated bright and dark vertical stripes. This produces a repeating pattern along the horizontal direction. First define a one-dimensional column signal within the window:

$$c_x(u) = \frac{1}{D} \sum_{z=0}^{D-1} I_{\sigma_z,\sigma_x}(z,u),$$

where $c_x(u)$ is the average intensity of column u. The local autocorrelation at lag k is

$$R_x(k) = \frac{\sum_u \left[c_x(u) - \bar{c}_x\right]\left[c_x(u+k) - \bar{c}_x\right]}{\sum_u \left[c_x(u) - \bar{c}_x\right]^2 + \varepsilon},$$

where:

- k is the horizontal lag in pixels,
- $\bar{c}_x$ is the mean column intensity in the window,
- $R_x(k)$ measures how closely the signal resembles a shifted copy of itself.

If a bright-dark pattern repeats every k pixels, the autocorrelation may show a peak near that lag. A periodicity score can be defined as

$$Q(x) = \max_{k \in \mathcal{K}} R_x(k),$$

where $\mathcal{K}$ is a clinically plausible range of stripe spacings. Additional periodicity descriptors may include:

- location of the strongest autocorrelation peak,
- height of the strongest peak,
- consistency of multiple peak spacings,

A high $Q(x)$ indicates repeated horizontal spacing of vertical structures.

6.11. Feature 4: Intensity Amplitude and Hypertransmission Strength

Barcode and atrophic regions may both appear brighter than normal tissue. Intensity amplitude is therefore useful but should not be used alone. The mean intensity in the window is

$$M(x) = \frac{1}{|W_x|} \sum_{(z,u)\in W_x} I_{\sigma_z,\sigma_x}(z,u).$$

A robust high-intensity measure may instead use an upper percentile (0.9) where $q_{0.90}(W_x)$ is the ninetieth percentile of intensities in the window. Possible amplitude features include:

- mean intensity,
- median intensity,
- ninetieth or ninety-fifth percentile intensity,
- fraction of pixels above a bright-pixel threshold,
- difference between upper and lower depth-region intensity.

Amplitude helps identify hypertransmission, but broad EA may also produce a high amplitude score. Therefore, amplitude should be interpreted jointly with verticality, persistence, and periodicity.

6.12. Feature 5: Spatial Heterogeneity

Spatial heterogeneity describes how much local intensity varies within the window. A barcode region may contain alternating bright and dark columns, whereas a broad EA region may be bright but comparatively smooth. The local variance is

$$H_{\text{var}}(x) = \frac{1}{|W_x|} \sum_{(z,u)\in W_x} \left[I(z,u) - \overline{I}_{W_x}\right]^2,$$

where $\overline{I}_{W_x}$ is the mean intensity within the window.

The local interquartile range and coefficient of variation may be defined as

$$H_{\text{IQR}}(x) = q_{0.75}(W_x) - q_{0.25}(W_x) \qquad H_{\text{CV}}(x) = \frac{\text{SD}(W_x)}{|\text{mean}(W_x)| + \varepsilon}.$$

Additional heterogeneity features may include:

- local entropy,
- gradient energy,
- peak density,
- variation in column persistence and intensity

A high heterogeneity score indicates substantial local variation, but it does not by itself establish that the variation is periodic or vertically aligned.

6.13. Feature Standardization

The raw features may have different numerical scales. For example, anisotropy may lie between 0 and 1, while variance may be much larger. Features must therefore be standardized before being combined. A robust standardized feature is

$$\widetilde{F}_j(x) = \frac{F_j(x) - \mathrm{median}(F_j)}{\mathrm{IQR}(F_j) + \varepsilon},$$

where:

- $F_j(x)$ is feature j at horizontal position x,
- $\mathrm{median}(F_j)$ is the median feature value across the scan,
- $\mathrm{IQR}(F_j)$ is the feature's interquartile range,
- $\widetilde{F}_j(x)$ is the standardized feature.

Robust standardization is preferred over ordinary mean-standard deviation scaling because abnormal regions may produce strong outlying values.

6.14. Weighted Barcode Score

The standardized features are combined into a continuous barcode score:

$$S(x) = w_V\widetilde{V}(x) + w_P\widetilde{P}(x) + w_Q\widetilde{Q}(x) + w_A\widetilde{A}(x) + w_H\widetilde{H}(x),$$

where:

- $\widetilde{V}(x)$ is standardized verticality,
- $\widetilde{P}(x)$ is standardized depth persistence,
- $\widetilde{Q}(x)$ is standardized periodicity,
- $\widetilde{A}(x)$ is standardized intensity amplitude,
- $\widetilde{H}(x)$ is standardized heterogeneity,
- w_V, w_P, w_Q, w_A, w_H are nonnegative feature weights.

For interpretability, the weights may initially be constrained to satisfy $\sum w_i = 1$. An initial proof-of-concept configuration could be

$$w_V = 0.25, \qquad w_P = 0.25, \qquad w_Q = 0.20, \qquad w_A = 0.15, \qquad w_H = 0.15.$$

These values are only starting values and do not represent validated clinical importance.

6.15. Choosing and Tuning Feature Weights

Weight selection should proceed in stages.

6.15.1. Stage 1: Equal or Expert-Specified Weights

The first implementation should use either equal weights or simple clinically motivated weights. This provides an interpretable baseline and allows inspection of the individual feature signals.

6.15.2. Stage 2: Grid Search Against Manual Annotations

The manually labeled Normal, EA, and Barcoding intervals can be converted into column-level reference labels. Candidate weight combinations may be evaluated using a constrained grid search. Each combination is evaluated by comparing the predicted barcode mask with the manual reference mask.

Possible performance measures include:

- sensitivity, specificity, precision:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- or an aggregate like F1 score, intersection over union, or area under the receiver operating characteristic curve.

$$F_1 = 2\frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}$$

The F1 score balances missed barcode regions and false-positive detections.

6.15.3. Stage 3: Learned Linear Weights

Once enough labeled scans are available, the weights can be estimated using logistic regression:

$$\Pr(Y(x) = 1) = \frac{1}{1 + \exp\left[-\left(\beta_0 + \sum_j \beta_j \widetilde{F}_j(x)\right)\right]},$$

where:

- $Y(x) = 1$ indicates a barcode-labeled column,
- β_0 is an intercept,
- β_j is the learned coefficient for feature j,
- the resulting value is the estimated probability of barcoding.

Logistic regression is useful because the learned coefficients remain directly interpretable. A positive coefficient means that larger values of the feature increase the predicted probability of barcoding.

Regularization may be used to reduce overfitting:

$$\mathcal{L}_{\text{penalized}} = \mathcal{L}_{\text{logistic}} + \lambda \sum_j \beta_j^2,$$

where:

- $\mathcal{L}_{\text{logistic}}$ is the logistic loss,
- λ controls the strength of regularization,
- larger λ discourages excessively large coefficients.

6.16. Threshold Selection

A horizontal position is initially classified as barcode-positive when $S(x) > \tau$ where τ is the barcode-score threshold. The resulting raw mask is

$$B_{\text{raw}}(x) = \mathbf{1}\left[S(x) > \tau\right].$$

Threshold-selection approaches include:

- maximizing the F1 score/Youden's Index $J = \text{Sensitivity} + \text{Specificity} - 1$.
- selecting a high percentile from normal scans,
- fitting a two-component mixture model,
- using a clinically specified sensitivity target.

The selected threshold is the value that maximizes J when equal importance is assigned to sensitivity and specificity.

6.17. Spatial Post-Processing

Thresholding may produce isolated positive columns or short gaps. The raw mask should therefore undergo simple one-dimensional spatial cleanup.

The post-processing steps are:

1. remove positive runs shorter than a minimum interval width,
2. fill negative gaps shorter than a maximum gap width,
3. convert the cleaned mask into contiguous horizontal intervals.

Initial configurable values may be

$$L_{\min} = 8 \text{ pixels}, \qquad G_{\max} = 4 \text{ pixels},$$

where:

- $L_{\min}$ is the minimum retained barcode interval width,
- $G_{\max}$ is the largest gap that may be filled between adjacent detections.

These values should be tuned against manual annotations and evaluated through sensitivity analysis.

6.18. Final Outputs

The detector should be able to return the preprocessed image, the normalized/denoised image, plots for the features and score, the mask overlay for detected barcoding, and barcoding metrics (width, number of intervals, min and max width)

The final visual display should contain:

1. the processed sub-BM scan,
2. highlighted vertical regions corresponding to detected barcode intervals,
3. the individual feature curves,
4. the combined barcode score,
5. the selected threshold.

6.19. Sensitivity and Validation Plan

The detector should be evaluated under reasonable changes to preprocessing methods, denoising parameters, and other preprocessing configurations.

Validation should be performed at three levels:

1. **Column level:** agreement between predicted and manually labeled horizontal positions.
2. **Interval level:** overlap between predicted and manually marked barcode regions.
3. **Scan level:** reproducibility across neighboring B-scans and across different OCT volumes.

Because neighboring columns and neighboring B-scans are not statistically independent, training and validation data should be split by subject or volume, not by individual image column. This prevents information from the same scan from appearing in both the training and evaluation sets.

6.20. Initial Recommended Proof-of-Concept Configuration

The initial detector configuration is:

- BM flattening,
- 150-pixel sub-BM crop,
- whole-ROI z-score normalization,
- anisotropic Gaussian denoising with $\sigma_z = 1.0$ and $\sigma_x = 0.5$,

- 21-pixel overlapping windows,
- stride of 1 pixel,
- full 150-pixel window depth,
- verticality, depth persistence, periodicity, amplitude, and heterogeneity features,
- robust feature standardization,
- interpretable weighted score,
- threshold-based classification,
- one-dimensional spatial cleanup.

This configuration is intended only as a reproducible starting point. Normalization, denoising, window dimensions, feature definitions, weights, and thresholds must be validated across multiple manually annotated OCT scans before being treated as final defaults.